BUSINESS CONTINUITY AND DISASTER RECOVERY

NAME: EBERE EMEILIA IKECHUKWU
STUDENT ID: GCR12707
OFFICIAL EMAIL: gcr2512707@students.icdfa.edu.ng
SUBMISSION DATE: 23/10/2026
CASE STUDY: NORTHBRIDGE SKILLS INSTITUTE

Executive Summary

This report explains how Northbridge Skills Institute can keep essential education and administration services operating after flooding and a power failure make its main site and local systems unavailable. The incident begins at 08:00 WAT on Monday (Day 0), and the site is expected to remain inaccessible for 72 hours. Microsoft 365 identity and email remain available through mobile internet, while local applications depend on AUTH01, DB01 and the relevant application servers. The report assesses business impact, sets proposed recovery objectives, reviews the backup design, compares recovery packages, and establishes coordinated continuity and disaster recovery procedures. It also models a recovery schedule using two technicians and documents decisions for a 45-minute tabletop exercise. The recommended starting strategy is a warm recovery site, supported by independent protected backups, subject to validation of actual performance. Temporary services must be approved, secure and reconciled after systems return. All cost and timing figures are scenario estimates, not actual losses or live test results.

Task 1: Business Impact Analysis

1.1 Purpose and approach

A Business Impact Analysis (BIA) identifies the services that matter most, the consequences of interruption and the resources needed to maintain an acceptable level of service. At Northbridge, business priority is not identical to technical restoration order. The examination is urgent because it starts at 11:00 but LMS01 cannot be restored until the network, AUTH01, security checks and DB01 are ready. The plan therefore needs both a continuity workaround and a technically feasible recovery sequence. Owners and titles used below are proposed roles for this fictional institute; they should be mapped to actual staff if the scenario is implemented.

1.2 BIA matrix

Function/Owner	Users Affected	Critical Resource and Dependencies	Impact Over Time	Minimum Acceptable Service and Controls	Priority
F1 Teaching and assessment — Academic Director / Examination Lead	Learners, lecturers, invigilators	LMS01, DB01, network, AUTH01, assessment materials, alternate venue, secure access; relies on staff and internet	Examination due at 11:00. Failure to provide an acceptable service triggers a one-off ₦120,000 rescheduling cost; longer disruption delays learning and may affect fairness and trust.	Use an approved assessment platform or controlled alternative session. Verify learner identity, attendance, timing and submissions; restrict access to papers and results.	1 — immediate
F2 Fees and finance — Finance Manager	Learners, finance staff, management	FIN01, DB01, payment provider, bank/payment records,	Fee confirmation due at 16:00 on Day 0. Missing local	Keep a restricted temporary register of provider references and	2 — same day

		network and AUTH01	records may cause wrong balances, duplicate posting or reconciliation delays.	receipts. Apply two-person review and reconcile before posting to FIN01.	
F3 Admissions and registration — Admissions Manager	Applicants, learners, admissions staff	REG01, DB01, network, AUTH01, application records and communications	Admissions close at 18:00 on Day 0. Lost or unrecorded applications and changes may disadvantage applicants or create disputes.	Use a timestamped, access-controlled intake register and acknowledgement numbers; obtain approval and reconcile each entry later.	3 — same day
F4 Staff payroll — HR/Payroll Manager	90 staff, HR and finance	HR01, DB01, network, AUTH01, approved payroll file and bank submission channel Bank submission deadline is 12:00 on Day 1. Incorrect or duplicate instructions could affect pay and require correction.	Admissions close at 18:00 on Day 0. Lost or unrecorded applications and changes may disadvantage applicants or create disputes.	Use the last approved payroll source, documented change control and independent verification. Do not submit unverified payment-file changes.	4 — before Day 1 deadline
F5 Records and certificates — Registrar / Records Manager	Learners, graduates, authorized requesters	FS01, network, AUTH01, protected files, access permissions and records staff	Requests have a 48-hour internal service commitment. Altered, lost or exposed files can harm privacy, trust and record integrity.	Log and acknowledge requests, prioritize urgent cases, verify requester authority and disclose only approved records. Reconcile activity after FS01 returns.	5 — within service commitment

1.3 Disruption cost calculations

The assignment says the hourly amounts are separate, non-overlapping incremental disruption costs. The 8-hour and 24-hour estimates assume no recovery or continuity measures. Because the examination is not available by 11:00 under that assumption, the one-off rescheduling cost is added once to F1.

Function	Hourly Rate	8-Hour Calculation	24-Hour Calculation
F1 Teaching and assessment	₦60,000	₦60,000 × 8 + ₦120,000 = ₦600,000	₦60,000 × 24 + ₦120,000 = ₦1,560,000
F2 Fees and finance	₦40,000	₦40,000 × 8 = ₦320,000	₦40,000 × 24 = ₦960,000
F3 Admissions and registration	₦20,000	₦20,000 × 8 = ₦160,000	₦20,000 × 24 = ₦480,000
F4 Staff payroll	₦10,000	₦10,000 × 8 = ₦80,000	₦10,000 × 24 = ₦240,000
F5 Records and certificates	₦5,000	₦5,000 × 8 = ₦40,000	₦5,000 × 24 = ₦120,000
Total	₦135,000 combined hourly rate	₦1,080,000 + ₦120,000 = ₦1,200,000	₦3,240,000 + ₦120,000 = ₦3,360,000

The estimates have at least two limitations. First, the hourly rates are supplied exercise assumptions and may not represent actual lost revenue, extra labour, supplier charges or recovery expenditure. Second, the calculation assumes costs accrue evenly over time and does not measure every indirect consequence. It does not fully value learner welfare, missed learning outcomes, reputational damage, privacy harm or loss of data integrity. These impacts remain important even when they cannot be expressed accurately in naira.

1.4 Priority and dependency risks

The examination has the strongest immediate deadline, so continuity arrangements must be in place before 11:00 even though the application cannot be restored by that time in the baseline schedule. Finance and admissions follow because their deadlines are 16:00 and 18:00 on Day 0. Payroll must be ready before the Day 1 noon bank deadline, while records and certificates must meet the 48-hour service commitment. Shared infrastructure creates several dependency risks. A network or AUTH01 failure blocks multiple functions; DB01 is a common dependency for F1–F4, so its delay affects four services; a failed incremental chain can prevent reliable restoration of several applications; the alternate facility has fewer workstations than the 40 learners who require them; and production and backup administration share an administrator identity, which may allow a compromised account to affect both environments.

Task 2: Recovery Objectives and Backup Design

2.1 Definitions and proposed objectives

Recovery Time Objective (RTO) is the target time within which a service should be restored after disruption. Recovery Point Objective (RPO) is the maximum acceptable data-loss interval, measured backwards from the disruption. Maximum Tolerable Downtime (MTD) is the longest period a business function can remain unavailable before its impact becomes unacceptable. The proposed targets below are planning commitments that require approval and testing. The MTD is set later than the RTO to leave a buffer for validation, business acceptance and unexpected delays.

Function	Proposed RTO	Proposed RPO	Proposed MTD	Reasoning
F1 Teaching and assessment	4 hours for LMS; continuity service by 11:00	1 hour for DB01	6 hours	The technical target is challenging; an approved alternative assessment service is essential because the exam begins three hours after disruption.
F2 Fees and finance	6 hours	1 hour for finance transactions	8 hours	Manual receipt capture and reconciliation protect the 16:00 deadline.
F3 Admissions and registration	8 hours	1 hour for registration changes	12 hours	Timestamped temporary intake protects applicants before the 18:00 close.
F4 Staff payroll	12 hours	1 hour for payroll changes	24 hours	Allows validation before the Day 1 bank deadline.
F5 Records and certificates	24 hours	4 hours for file changes	48 hours	The MTD reflects the internal service commitment; access control and file integrity remain critical.

The targets are proposed, not guaranteed service levels. The modelled warm-site schedule reaches LMS01 at 11:45, three hours and 45 minutes after the disruption, within a four-hour RTO by only 15 minutes. The examination deadline is nevertheless missed, so the continuity service must be treated separately from technical recovery. The payroll target is measured from the start of the incident, while the bank submission deadline is on Day 1; that time must be used for verification, not as a reason to relax payroll controls.

2.2 Backup Checkpoint Age

Checkpoint	Timestamp	Age at 08:00	Implication
DB01 full backup B01	23:00 Day −1	9 hours	Base backup only; it is not the latest point by itself.
DB01 incremental chain B02	07:00 Day 0	1 hour	Latest recoverable business-data point if B01

			and every required incremental are valid.
AUTH01 snapshot B03	00:00 Day 0	8 hours	Authentication/configuration state may be eight hours behind.
FS01 full backup B04	23:00 Day −1	9 hours	Records and certificate files may be nine hours behind.

DB01's one-hour potential data-loss window matches the proposed one-hour RPO only if the full backup and every required incremental pass integrity check and restore consistently. AUTH01's eight-hour age exceeds its proposed one-hour RPO. FS01's nine-hour age exceeds the proposed four-hour RPO. Temporary receipts, payment-provider references, admission acknowledgements and approved payroll change logs can help reconstruct documented transactions after a checkpoint. They cannot automatically recover deleted files, unrecorded changes, overwritten information or the exact database state without evidence.

2.3 Improved Backup Design

Northbridge should take a full DB01 backup each night and incremental backup at least hourly. Every incremental should have a timestamp, identifier and manifest. To restore DB01 to 07:00 on Day 0, the team must restore B01 from 23:00 on Day −1 and then apply every required incremental in the chain through 07:00. Missing a required link may make the intended recovery point unusable. AUTH01 configuration and directory-state backups should be protected at least hourly. FS01 should receive incremental or differential backups at least every four hours, with more frequent protection for especially critical records if feasible. A proposed retention baseline is daily recovery points for 30 days, weekly copies for 12 weeks and monthly copies for 12 months, subject to legal, privacy and records-retention requirements.

The 3-2-1 principle means keeping at least three copies of important data, on two types of storage, with at least one copy offsite. One copy should be offline or immutable and managed with credentials separate from production. Copies should be encrypted in transit and at rest, with keys protected separately. Backup administrators should use unique accounts, multifactor authentication, least privilege and logged access. The disconnected full backup from Day −3 provides another recovery option, but it is older and should be used only if newer recovery points fail or are untrustworthy.

Offsite storage does not prove that a backup can be restored. The team should verify checksums or integrity results, confirm backup access is independent from production credentials, and restore copies in an isolated environment. Testing should include database consistency, application log-in, permissions, sample transactions, file readability and business-owner acceptance. The assignment states that baseline checkpoints pass integrity checks, but offsite access independence is undocumented. It must be verified rather than assumed.

Task 3: Recovery Strategy and Budget

3.1 Compare the packages

Package	Base Annual Cost	First Critical Service Estimate	Data-Loss Window	Strengths and Limitations
Cold	₦600,000	24–48 hours	Up to 24 hours	Lowest base cost, but too slow for the examination and same-day deadlines;

				requires more setup after activation.
Warm	₦1,500,000	3–5 hours	1–6 hours	Middle-ground option. The drill reaches LMS01 in 3h45, but backup RPO gaps and deadline workarounds remain.
Hot	₦2,400,000	About 1 hour	Up to 15 minutes	Fastest stated recovery and smallest data-loss window but uses most of the budget and still needs protected backups and validation.

3.2 Complete Annual Budget

Package	Base Cost	Protected Backups	Training and Tests	Annual Total	Budget Headroom
Cold	₦600,000	₦400,000	₦200,000	₦1,200,000	₦1,800,000
Warm	₦1,500,000	₦400,000	₦200,000	₦2,100,000	₦900,000
Hot	₦2,400,000	₦400,000	₦200,000	₦3,000,000	₦0

All three packages fit the ₦3,000,000 limit when both mandatory additions are included. I recommend the warm package as the initial planning choice because it leaves ₦900,000 in headroom and can be exercised with the two-technician team. This recommendation is conditional on verifying the provider's actual service levels, connectivity, facility location and support arrangements. Cold is cheaper but does not fit the immediate needs. Hot has the best stated recovery window but consumes the full budget, leaving no headroom.

The warm package cannot automatically be said to meet every objective. Its estimated first-service window is 3–5 hours and its stated data-loss window is 1–6 hours. The modelled drill reaches LMS01 after 3h45, but the 11:00 examination still needs a workaround and a six-hour data-loss window does not meet a one-hour RPO unless backup frequency and restore capability are improved. Residual risk should be accepted in writing by the executive sponsor after review by business owners and the risk/security lead. If testing shows the warm package cannot meet approved objectives, management should reconsider hot recovery or formally approve revised targets based on business impact.

The estimated disruption cost is ₦1.2 million for eight hours and ₦3.36 million for 24 hours. This comparison shows that one serious incident could cost more than the warm package's annual price, but it does not prove a positive annual return on investment. ROI requires outage probability and frequency, expected avoided loss, implementation and operating costs, and evidence of how much controls reduce incident likelihood or severity. Before approval, management should request supplier service-level evidence, actual recovery test results, facility and connectivity details, data-protection terms, exit arrangements, backup access-control evidence and a clear list of included and excluded costs.

Task 4: Coordinated Continuity and Disaster Recovery Plan

4.1 Scope, activation and command

The plan covers F1–F5, shared infrastructure, staff, suppliers, communications and the temporary operating site. The Incident Lead, normally a designated senior manager, may activate the plan when the main site or essential systems are unavailable and the disruption threatens safety, a critical deadline, data integrity or an approved RTO. A named deputy takes over if the Incident Lead cannot be reached. First actions are to protect people, confirm site safety with Facilities, notify business owners, open an incident log, establish a communication channel through Microsoft 365 and mobile internet, and issue situation reports every 30 minutes during the initial response.

The continuity team coordinates temporary services and learner support. The DR team restores technology and validates backups. Facilities confirms whether the building and utilities are safe. The Security/Incident Response Lead investigates suspicious access, unexpected changes, failed integrity checks or possible unauthorized activity. Flooding and power failure alone do not prove a cyberattack. However, suspicious backup-administrator activity or unexplained corruption must be investigated before affected backups are trusted.

4.2 RACI matrix

Activity	Incident Lead	DR Lead	Business Coordinators	Facilities Lead	Security/IR Lead
Activate plan	A	C	R	C	I
Arrange temporary examination	A	C	R	C	I
Validate backup and access independence	I	A/R	C	I	C
Restore network, AUTH01 and applications	I	A/R	C	I	C
Investigate possible compromise	I	C	I	I	A/R
Communicate with learners and staff	A	I	R	C	C
Approve return to primary site	A	C	C	R	C

A means Accountable, R Responsible, C Consulted and I Informed. Each activity has exactly one accountable role. The Incident Lead is accountable for overall activation and business communication, the DR Lead for technical restoration and the Security/IR Lead for compromise investigation.

4.3 Temporary Procedures for Each Function

F1-Teaching and assessment: The Academic Director should approve the alternative assessment method before use. Eighty learners can use their own devices, but the 40 learners requiring centre workstations cannot all be seated at the initial venue, which has only 25 workstations. The institute should seek an additional approved venue or arrange a controlled alternative session for the shortfall of 15. Arrangements must protect fairness, accessibility, learner identity, invigilation, question confidentiality and the audit trail for attendance and submissions. Learners must not be sent to an unsafe or unapproved location.

F2-Fees and finance: Finance staff should record each payment reference, amount, timestamp and payer identifier in a restricted temporary register. A second authorized person should review the register daily. The payment provider remains available, but a successful payment must not be posted twice when FIN01 returns. Reconciliation should compare provider settlement data, temporary receipts and restored records. Unresolved differences should be escalated to the Finance Manager.

F3-Admissions and registration: Admissions should use a timestamped register, issue each applicant a reference number and communicate the temporary process through approved channels. Access should be limited to authorized staff. Sensitive documents must not be sent through personal email or unapproved messaging services. After REG01 is restored, each record should be entered once and independently checked.

F4-Payroll: Payroll staff should use the last approved payroll record and a documented change log. Two authorized people should verify changes, totals and bank-file details before submission. If the data cannot be verified, the Payroll Manager should escalate before the deadline rather than send uncertain or duplicate instructions. Temporary changes must be reconciled with HR01.

F5-Records and certificates: Records staff should log each request, verify requester identity and authority, and provide only the minimum necessary information. Requests should be prioritized by urgency while respecting the 48-hour service commitment. Temporary copies must be kept in an approved, access-controlled location, and all issued records reconciled with FS01 after restoration.

4.4 Technical Recovery Runbook

Step / Owner	Prerequisite	Completion Check	If the step fails
1. Confirm safety and activate- Incident Lead / Facilities	Site outage confirmed	Incident ID, decision log and notifications complete	Prohibit entry if safety is uncertain; coordinate remotely.
2. Prepare recovery network-DR Technician 1	Approved recovery environment	Connectivity, segmentation and logging checked	Use approved alternate connectivity and escalate provider failure.
3. Restore AUTH01 - DR Technician 1	Network ready	Directory services and configuration validated	Use known-good snapshot; isolate if access looks suspicious.
4. Validate security and backups - DR Lead / Security Lead	AUTH01 restored	Integrity, chain completeness,	Do not use a failed or suspicious chain; preserve evidence

		permissions and access logs reviewed	and select a verified point.
5. Restore DB01-DR Technician 1	Checks passed	Database consistency and sample records checked	Stop and select a known-good point if validation fails.
6. Restore LMS01, FIN01, REG01 and HR01-both technicians	DB01 validated	Log-in, sample transactions and role permissions tested	Prioritize based on business need and prerequisites.
7. Restore FS01-DR Technician 2	AUTH01 and security checks complete	Sample files readable and access restrictions verified	Use a verified copy and record stale or missing files.
8. Business acceptance-relevant owners	Technical checks passed	Owners sign acceptance: temporary records reconciled	Keep workaround active if the service is not fit for use.
9. Controlled return-Incident Lead / Facilities	Main site cleared; systems validated	Transfer log, reconciliation and final approval complete	Remain at alternate site until safety and service acceptance are documented.

Restoring a server is not the same as restoring an acceptable business service. Business owners must confirm that the service is accurate, accessible to the right users and fit for purpose. Return to the main site requires written Facilities clearance, verified power and environmental conditions, secure network readiness, controlled transfer of new records and final sign-off. Temporary records must be reconciled before duplicate entry or conflicting versions can occur.

4.5 Communication Matrix

Audience	**Timing**	**Channel**	**Owner**
Learners and applicants	Initial notice; then at least every two hours or after a material change	Official email, website, approved SMS and LMS if available	Communications / Business Coordinator
Staff and lecturers	Initial instructions and scheduled updates	Microsoft 365, mobile calls and approved staff channel	Incident Lead
Senior management	Initial summary and 30-minute updates during active recovery	Approved incident channel and written situation report	Incident Lead

Suppliers and payment provider	When service status, recovery support or reconciliation requires it	Verified supplier contacts and support tickets	DR Lead / Finance Manager
Regulator or authority, if required	When legal, contractual or safety obligations require notification	Approved formal channel	Designated compliance/legal role

4.6 Initial Learner Notice

Subject: Temporary arrangements following today's site closure

Northbridge Skills Institute has temporarily closed its main site following flooding and a power failure. The safety of our learners and staff is our first concern. The site is currently expected to remain inaccessible for approximately 72 hours, but we will provide updates if this changes.

The examination scheduled for 11:00 is being managed under an approved continuity arrangement. Learners should check their official institute email and other official communication channels for instructions about their assessment method, venue or any alternative session. Please do not travel to the closed site unless the Institute specifically instructs you to do so. Learners who require workstation access or accessibility support should contact the designated learner support team through an official channel.

Please keep any assessment or payment reference numbers you receive and do not send personal documents through unofficial accounts. We will provide the next update as soon as arrangements are confirmed. We apologise for the disruption and appreciate your patience while the Institute works to maintain essential services safely.

Possible compromise should trigger Security/Incident Response involvement when there is an unexpected, privileged login, suspicious change, malware indicator, failed integrity check or evidence of unauthorized access. The security lead should preserve logs, isolate affected systems where appropriate and investigate before suspect backups are used. Continuity services can continue through safe alternatives while the investigation proceeds.

Task 5: Recovery Schedule and Exercise

5.1 Modelled baseline schedule

The schedule starts at 08:00 WAT on Day 0. Each action occupies one technician for its full duration, and parallel work is permitted only after prerequisites are complete and a technician is free. The following is a modelled baseline, not a live recovery test.

Action	Technician	Start	Finish	Prerequisite
Recovery network	Technician 1	08:00	08:30	None
AUTH01 restoration	Technician 1	08:30	09:00	Recovery network
Security and backup checks	Technician 1	09:00	09:30	AUTH01 restoration
DB01 restoration	Technician 1	09:30	11:00	Security and backup checks

FS01 restoration	Technician 2	09:30	10:30	AUTH01 and security/backup checks
LMS01 restoration	Technician 1	11:00	11:45	DB01 restoration
HR01 restoration	Technician 2	11:00	11:45	DB01 restoration
REG01 restoration	Technician 1	11:45	12:15	DB01 restoration
FIN01 restoration	Technician 2	11:45	12:15	DB01 restoration

The assessment application's dependency path is network (30 minutes) → AUTH01 (30) → security and backup checks (30) → DB01 (90) → LMS01 (45). Total elapsed time is 225 minutes, so LMS01 is available at 11:45, three hours and 45 minutes after disruption. FS01 finishes at 10:30. HR01 finishes at 11:45, while REG01 and FIN01 finish at 12:15. All five functions are technically restored by 12:15 under this model. The 11:00 examination still requires a continuity workaround.

5.2 Compare Recovery Results with Objectives

Function	Modelled Finish	Elapsed Time	RTO / MTD	Assessment
F1 / LMS01	11:45	3h 45m	4h / 6h	Within RTO by 15 minutes, but the exam deadline needs a workaround.
F2 / FIN01	12:15	4h 15m	6h / 8h	Within RTO; temporary payment records must be reconciled.
F3 / REG01	12:15	4h 15m	8h / 12h	Within RTO; protect timestamped application intake.
F4 / HR01	11:45	3h 45m	12h / 24h	Within RTO; payroll approval and bank deadline controls remain necessary.
F5 / FS01	10:30	2h 30m	24h / 48h	Within RTO; its nine-hour-old checkpoint exceeds the proposed four-hour RPO.

DB01's latest recoverable point is one hour before disruption and matches the proposed one-hour RPO only if the incremental chain is valid. AUTH01 is eight hours old, exceeding its one-hour target. FS01 is nine hours old, exceeding its four-hour target. Timings do not prove that data-loss objectives are met; backup frequency and restore capability must be improved and tested. The Day 0 examination, finance and admissions deadlines still require continuity arrangements even when the technology recovers within RTO.

5.3 Forty-Five-Minute Tabletop Exercise

Minute / Inject	Owner	Decision and Revised Action	Expected Effect
10-Site closure extends from 72 hours to five days	Incident Lead	Extend alternate-site arrangements, staffing, learner support, supplies and supplier coverage; reassess daily.	Longer continuity period; higher cost and fatigue risk.
20-Alternate facility has only 15 working workstations	Academic Director / Facilities Lead	Recalculate capacity; seek another approved venue or controlled alternative session and communicate decisions.	More learners need an alternative; accessibility and fairness must be maintained.
30 -DB01 incremental fails integrity check; unexpected backup administrator login	DR Lead / Security Lead	Stop using the failed chain, preserve logs, investigate the account, validate an independent copy and select a known-good recovery point.	Recovery may be delayed and RPO missed; keep safe manual services running.
40-Payment provider reports successful payments missing from restored finance records	Finance Manager	Obtain provider transaction report, prevent duplicate posting, reconcile with temporary receipts and require second-person approval.	Finance remains provisional until discrepancies are resolved.

A facilitator can spend five minutes briefing participants, work through the four injects and decisions, and reserve the final ten minutes to review lessons. Each decision should be recorded with its time, owner, rationale and effect on service or recovery. The injects are a stress scenario and must be kept separate from the baseline schedule.

5.4 Corrective-Action Register

Priority	Action	Owner	Target	Measurable evidence
High	Separate backup administration from production identity; enforce MFA and	Security Lead	30 days	Separate accounts confirmed; access test and reviewed logs retained.

	privileged access review			
High	Automate DB01 incremental-chain integrity checks	DR Lead	30 days	Successful alert test and documented failed-chain procedure.
High	Increase AUTH01 protection frequency and test restoration	Infrastructure Lead	45 days	Restore evidence shows checkpoint age of one hour or less.
High	Increase FS01 backup frequency	Records / IT Owner	45 days	Successful restore shows recovery point within four hours.
High	Approve examination continuity and workstation-capacity procedure	Academic Director	30 days	Signed procedure covers the 40 learners and the 15-seat shortfall.
Medium	Run a two-technician recovery exercise	DR Lead	90 days	Report compares actual times with RTOs and records lessons.
Medium	Reconcile payment provider records against FIN01	Finance Manager	Quarterly	Signed reconciliation explains or escalates all differences.

Task 6: Governance and Performance Measures

6.1 Ownership, Maintenance and Testing

The executive sponsor should approve the continuity policy, recovery objectives, budget and any accepted residual risks. A Business Continuity Coordinator should maintain the plan, while each function owner approves the temporary procedure for their service. The DR Lead maintains technical runbooks and dependencies. The Security/Incident Response Lead reviews privileged access, backup integrity and evidence preservation. Review the plan at least annually and after a major system or supplier change, a site move, an incident, a failed exercise, or a material change in learner numbers or service deadlines.

A practical annual calendar includes quarterly review of contacts and dependencies; quarterly sample restoration of critical files or database records; twice-yearly tabletop exercises; one annual end-to-end recovery exercise involving both technicians and business owners; annual staff awareness training; and a post-incident review after every real disruption. Training and test results should be retained, findings assigned owners and due dates, and the approved

plan distributed to authorized staff in a secure location that remains accessible if the main site is unavailable.

6.2 Framework Mapping

Assignment Output	NIST SP 800-34 alignment	ISO 22301 theme
BIA and priority matrix	Conduct a BIA and identify critical processes	Business impact analysis, risk assessment and continuity objectives
Recovery objectives and backup strategy	Identify preventive controls and create contingency strategies	Planning for disruption, resources and continuity capability
DR runbook and RACI	Develop the information-system contingency plan	Roles, responsibilities, operational planning and documented procedures
Tabletop exercise and corrective actions	Testing, training and exercises	Exercising, performance evaluation and continual improvement
Governance calendar, version control and KPIs	Plan maintenance	Monitoring, review, maintenance and improvement

This is an educational, high-level alignment, not a claim of certification or a full clause-by-clause compliance assessment. NIST SP 800-34 Rev. 1 describes a contingency planning process that includes policy, BIA, preventive controls, recovery strategies, a contingency plan, testing/training/exercises and maintenance. ISO 22301 provides a management-system framework for planning, implementing, monitoring, reviewing, maintaining and continually improving business continuity.

6.3 Flood and Climate Consideration

Flooding can prevent physical access even when technology is recoverable. Northbridge should review drainage, equipment placement, power protection, flood history, evacuation routes and the location of its alternate facility. The alternate site should not share the same flood exposure or a single vulnerable power or network route with the main site. Climate-related disruption should be included in the annual risk assessment and supplier review, consistent with the climate amendment referenced in the assignment.

6.4 Key Performance Indicators

KPI	Formula / Method	Data Source	Target	Frequency / Owner
Recovery objective achievement	Services restored within approved RTO ÷ total services tested × 100	Exercise report and timestamps	At least 90% initially; 95% by month 12	After exercises and quarterly — DR Lead

Restoration-test coverage	Systems successfully restore-tested ÷ systems in scope × 100	Backup and restoration register	100% of critical systems annually	Quarterly — Infrastructure Lead
Corrective-action closure	Actions closed by due date ÷ actions due × 100	Risk/action register	At least 90% on time	Monthly — Business Continuity Coordinator
Staff training completion	Staff completing assigned training ÷ staff assigned × 100	Training attendance/LMS records	At least 95% annually	Quarterly — HR/Training Lead
Backup integrity success	Backup jobs passing integrity checks ÷ jobs tested × 100	Backup console and independent test logs	At least 98%; every critical chain tested	Monthly — DR Lead

6.5 Maturity and 12-Month Improvement Target

The assignment refers to the maturity model on slide 13 of the ICDFA teaching deck. The slide's levels are not reproduced in the assignment text, so the labels should not be invented. Northbridge should assess itself using that exact model after reviewing the slide. The scenario nevertheless shows weaknesses: offsite backup access independence is undocumented, production and backup environments share an administrator identity and the AUTH01 and FS01 checkpoint ages exceed the proposed RPOs. A realistic 12-month objective is to advance by one level in the course model, supported by evidence of approved ownership, documented procedures, independent protected backups, successful restore tests, completed training and timely closure of corrective actions.

Conclusion

The flooding and power failure demonstrate why continuity planning must protect both technology and the services people rely on. Northbridge should prioritize the examination and same-day deadlines, maintain controlled temporary processes, restore shared dependencies in the correct order and reconcile manual records before declaring normal service. The warm package is a cost-conscious starting recommendation, but it must be validated against approved recovery objectives and supported by independent protected backups. Regular exercises, clear ownership and measurable corrective actions will help turn the plan into a usable capability rather than a document consulted only after an incident.

References

- *International Cybersecurity and Digital Forensics Academy (ICDFA). (n.d.). GRC205 Incident Management and Business Continuity, Week 21: Business Continuity and Disaster Recovery Assignment [Course assignment].*
- *National Institute of Standards and Technology. (2010). Contingency Planning Guide for Federal Information Systems (NIST Special Publication 800-34 Rev. 1). https://csrc.nist.gov/pubs/sp/800/34/r1/upd1/final*
- *International Organization for Standardization. (2019). ISO 22301:2019 Security and resilience — Business continuity management systems — Requirements. https://www.iso.org/standard/75106.html*
- *International Organization for Standardization. (2024). ISO 22301:2019/Amd 1:2024 — Climate action changes. Available through https://www.iso.org/standard/75106.html*